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Epileptic seizure detection by learning EEG as images using vision transformers

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ABSTRACT

The detection of epileptic seizures is crucial for anticipating and effectively managing seizure events. The accuracy of seizure detection from EEG signals effectively depends on the model's ability to capture the temporal characteristics within and across different types of signals. To address it, we first transform EEG data streams into different image representations to extract deeper features across signals and then leverage recent vision transformer models, such as Mix-ViT and Swin Transformer networks, to underscore the importance of combining various image representations to enhance the model's ability to detect seizures. Our approach uses diverse image representations for EEG sequence analysis, demonstrating that learning-based fusion of different image representations enhances seizures detection accuracy. The models proposed in this work were thoroughly evaluated on the Bonn University and CHBMIT datasets, where the experimental results revealed improvements upon many recent state-of-the-art methods. In particular, with the Bonn University dataset using the Swin-ViT Transformer, in a 10-fold cross-validation setting an accuracy of 100% was reported. For the CHB-MIT dataset, the results achieved 96.34% accuracy, 94.2% sensitivity, 97.8% specificity, 97.059% precision, 95.65% F1 score, and 98.96% AUC. These outcomes surpass other state-of-the-art methods, emphasizing the importance of combining diverse image representations for EEG signals.

Introduction

Epileptic seizure detection is a critical area of concern in the field of medical diagnostics, playing a pivotal role in the timely and accurate identification of seizures, enabling healthcare professionals to provide prompt medical intervention to patients suffering from epilepsy. According to the World Health Organization, epilepsy has a diagnosis rate ranging from 4 to 10 per 1000 individuals [1]. The importance of efficient seizure detection cannot be overstated, as it directly impacts the quality of life of individuals with epilepsy and ensures their safety during seizure. However, the task of detecting seizures remains a significant challenge in the medical community, with healthcare professionals often struggle to achieve consistent and reliable results. Being able to detect seizures in the preictal stage (stage before seizures) could avoid patients with epilepsy a serious injury which could lead as well to death [1,2].

Traditionally, epileptic seizure detection has relied on a combination of clinical observations, electroencephalogram (EEG) recordings, and expert analysis. While these methods have been valuable, they are inherently limited by the subjectivity of human explanation and the need for continuous monitoring. This often leads to missed or delayed detection, potentially risking the well-being of patients. Furthermore, numerous smart devices are available in the market designed to assist in identifying seizures. Therefore, the automatic detection of seizures using smart devices is imperative to provide a more practical and comfortable solution for patients [2].

In recent years, there has been a growing interest in harnessing the power of machine learning approaches to enhance the accuracy and efficiency of epileptic seizure detection [3]. Many studies were made to diagnose EEG using machine learning techniques. Some researchers worked on it as computer vision problem where they converted the signals into images [4], and others used basic machine learning techniques such as support vector machine (SVM), decision tree and random forest [5]. These machine learning methods leverage algorithms and models to analyze the vast amounts of data generated by EEG and other physiological sensors, providing healthcare professionals with timely alerts and insights into a patient's condition. Attention mechanisms, particularly transformer architectures, enable models to focus on relevant information and relationships within input sequences, leading

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to more accurate predictions. Transformers have gained considerable attention in various domains, including seizure detection, owing to their effectiveness [6,7].

Additionally, one notable method for detecting seizures involves converting EEG time series data into images, a process known as time series to image conversion [8]. By representing EEG signals as images, researchers can utilize image processing techniques to extract significant features related to seizure events. Each image representation captures distinct features of the input data, highlighting the importance of selecting the most suitable representation for capturing the key features of an EEG signal. It is important to note that the choice of image representation is not the only factor influencing model performance; distinct models may show disparate performance with different representations. This variability arises from the ability of each model to capture different features, with some models potentially finding certain features more valuable than others in a specific image representation. Moreover, different datasets belonging to the same problem may also exhibit varied performance when using the same image representation and model, emphasizing the need for thorough evaluation across different datasets.

Despite the existing studies in the field and the contributions made by researchers in detecting seizures, seizure detection remains an unsolved issue that requires further enhancement. Finding a method to improve the performance of seizure detection is still an open area of research. Furthermore, image representations for time-series sequences can capture important features in EEG signals. In this context, the novelty of our work is summarized in the following points:

1. **Vision Transformer for Time-Series Analysis:** Recent advancements in vision transformer models, namely Mix-ViT [9] and Swin [10], have been recognized for their remarkable performance on image classification problems. In this work, we present a novel application of Mix-ViT and Swin transformer models for seizure detection in time series sequences.
2. **Learning Based Fusion of Time-Series Image Representation:** Further, the next major contribution of this work is the application of learning-based combinations of image representations – such as spectrograms, scalograms, and raw images – using both Swin and Mix-ViT vision transformer models to enhance performance by extracting critical features for seizure detection from EEG signals from different image representations.
3. **Thorough Evaluation and Comparative Analysis:** Extensive testing validates the robustness of the approach under various conditions. Additionally, models are compared against recent state-of-the-art approaches demonstrating superior performance in seizure detection tasks on two widely used public datasets, namely Bonn University [11] for multiclass classification and CHB-MIT [12] for binary classification. It was observed that the learning-based fusion of various types of image representations with transformer networks can significantly improve detection performance.

The rest of the paper is organized as follows: Section "Literature review" provides an overview of recent advancements in the field. Section "Proposed system for EEG-based seizure detection" introduces the proposed framework. Section "Methods employed in the proposed system" explains the underlying techniques and approaches. Section "Methodology for experimental evaluation" presents the experimental setup and performance analysis. Finally, Section "Conclusion" summarizes the findings and highlights future research directions.

Literature review

This section explores some important research methods in the field of machine learning and seizure detection, highlighting traditional approaches as well as new methods for identifying seizures.

Machine learning methods for EEG sequence analysis

Seizure detection research has evolved significantly, starting with classical machine learning approaches like SVM and progressing to advanced deep learning models. In 2010, Shoeb et al. used SVM for seizure classification, paving the way for the adoption of deep learning methods [13]. Convolutional Neural Networks (CNNs) have shown remarkable success in classifying epilepsy [3,14,15], achieving up to 99.83% accuracy on the Bonn University dataset by combining CNNs for feature extraction with classical classifiers [14]. Additionally, preprocessing techniques have proven to enhance results. Kavitha et al. used discrete wavelet transform (DWT) and statistical features with various classifiers [16], while Assali et al. employed sample entropy preprocessing with a 1D CNN [17], achieving 90.1% accuracy on the CHB-MIT dataset.

Classical methods, such as SVM with empirical mode decomposition, showed high sensitivity (93%) on the CHB-MIT dataset [18], but combining CNNs with Long Short Term Memory (LSTM) models yielded further improvements [19]. Furthermore, LSTM networks have been effective in handling noisy data [20], with approaches integrating CNNs and LSTM/Bi-LSTM networks achieving accuracy rates as high as 98.44%–100% on datasets like Bonn University [21].

The field is now moving towards sophisticated architectures, including time series-to-image conversion, residual connections, transformers, and vision transformers, promising further advancements in seizure detection.

Analysis of EEG signals using image representations

Transforming time series data into visual representations, particularly for EEG signals, offers significant potential across various fields. Techniques like recurrence plots [8] and Gramian Angular Summation Field (GASF) [22] are foundational for machine learning models such as CNNs and LSTMs. Similar advancements in feature extraction and AI-driven analysis have also been explored in other domains, such as image steganalysis [23,24] and intelligent edge computing for IoT systems [25], demonstrating the broader applicability of deep learning for complex data interpretation.

Recurrence plots have been widely used for capturing EEG signal features. Studies like those by Zeng et al. [8], Shankar et al. [22], and Ravi et al. [26] demonstrate the effectiveness of applying recurrence plots with different neural network models on datasets such as Bonn and CHB-MIT. Additionally, modified recurrence plots by Huang et al. [27] achieved superior classification performance, validating the utility of recurrence-based techniques in EEG signal analysis.

Alternative image representations, including GASF transformation, Continuous Wavelet Transform (CWT), Discrete Wavelet Transform (DWT), and Poincaré plots, are well-established methods for capturing both temporal and spatial features. These techniques are widely employed due to their effectiveness in enhancing the analysis of time series data. Shankar et al. combined GASF transformation with CNNs on the Temple University Hospital dataset, achieving 96.01% accuracy for two classes of seizures [22]. Varlıand Yılmaz utilized CWT and Short-Time Fourier Transform (STFT) to extract features, proposing two base models (2DCNN CWT-LSTM and 2DCNN-STFT-LSTM), with the latter performing better on various datasets [28].

Poincaré plots, used for seizure detection by Akbari et al. [29], employed DWT and a binary model, achieving results comparable to other state-of-the-art methods. Researchers are also exploring transfer learning (TF) in EEG data classification, enhancing accuracy through pretrained architecture [29]. Goel et al. developed an epilepsy classifier using recurrence plots and ResNet-50, achieving 95.34% accuracy on the Bonn University dataset [30]. Ilias et al. proposed models using pretrained networks and STFT for image construction, yielding promising results [4].

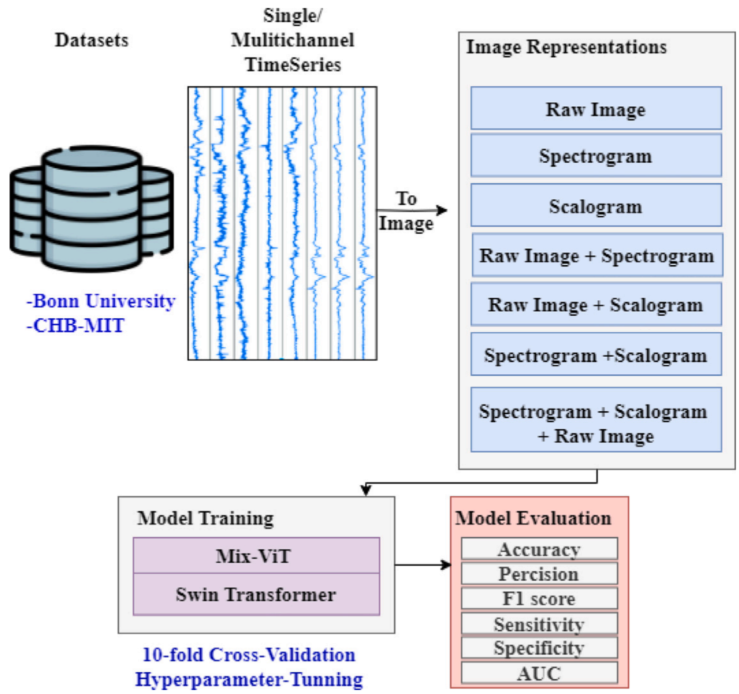

Fig. 1. Working strategy.

Different image representations of EEG signals, such as recurrence plots, GASF, wavelet transforms, and Poincaré plots, enhance seizure detection by capturing various features. Each method highlights distinct aspects—non-linearities, frequency content, and dynamic behaviors—allowing machine learning models to identify complex seizure patterns effectively and improve classification accuracy across datasets.

Transformers for epileptic seizure detection using EEG signals

While transformers excel in capturing temporal information and managing dependencies, their application in vision tasks opens up new avenues for research. This section focuses on Vision Transformers (ViTs) and their unique attributes, which can significantly enhance visual analysis, particularly in EEG data. ViTs have recently gained traction in this area, with early works demonstrating their superiority over traditional CNNs in image classification tasks [31]. However, the performance of ViTs is sensitive to dataset size, lacking the local inductive bias of CNNs [32]. This characteristic impacts their effectiveness on smaller datasets but offers potential when trained on larger collections.

In the domain of epileptic seizure detection, the application of Vision Transformers (ViTs) is still in its early stages. Zhang et al. [33] introduced a personalized seizure prediction model that leverages ViTs to analyze scalp EEG signals from the CHB-MIT dataset. While the details of the ViT architecture were not extensively described, the authors employed Leave-One-Out Cross-Validation (LOOCV) to thoroughly evaluate their model. Hussein et al. [34] expanded on this work by utilizing CWT to convert EEG data into image-like representations for a multichannel ViT model, achieving promising results across three datasets.

Godoy et al. [35] introduced a convolutional ViT model that leverages multichannel EEG data, detecting temporal features without preprocessing. This model performed exceptionally well on the CHB-MIT dataset and outperformed both standalone CNNs and ViTs by using raw EEG signals as input. Table 1 summarizes techniques using ViT for EEG analysis.

Despite the potential of ViTs in seizure detection, standardization and deeper insights into their application are crucial for further advancement. Recent variations of ViTs, like Mix-ViT [36], integrate multi-attention mechanisms for ultra-fine-grained categorization, enhancing performance while using fewer parameters. Additionally, Li et al. [10] introduced ViTST, which transforms irregularly sampled time series into line graphs, effectively combining local and global feature extraction. This method demonstrates robustness in healthcare applications, highlighting the versatility of ViTs in various domains.

Proposed system for EEG-based seizure detection

The working strategy, illustrated in Fig. 1, begins by converting time-series data from the Bonn University and CHB-MIT datasets into various image representations, including raw images, spectrograms, and scalograms. These image representations are then used as inputs for two deep learning models, Mix-ViT and Swin Transformer, to investigate the impact of different image types on model performance. To ensure a thorough evaluation, seven different combinations of image representations were tested using a robust 10-fold cross-validation process. This iterative testing process was designed to determine which combination of image type and model yields the best results. Following the processing by these models, the results are systematically analyzed and evaluated using six key metrics: accuracy, precision, F1-score, sensitivity, specificity, and Area Under the Curve (AUC). This comprehensive approach highlights the effectiveness of leveraging image-based representations to enhance the analysis of complex time-series data.

Hyperparameter tuning and optimization

The fine-tuning process for the Mix-ViT and Swin networks focused on optimizing key hyperparameters, including dropout rate, patch size, number of transformer layers, and learning rate. A grid search strategy was employed to tune these hyperparameters, exploring learning rates [$1e^{-2}$–$1e^{-5}$], dropout rates [0.3–0.6], patch sizes [8–32], and transformer layers [6–12]. Experiments were conducted to determine the optimal values for these parameters, ensuring improved performance across metrics such as sensitivity, precision, F1 score, AUC, and accuracy. This systematic exploration was essential for configuring the models effectively for robust feature extraction and accurate seizure detection.

Table 1
Summary of EEG seizure detection studies utilizing ViT.

Ref.	Preprocessing	Model used	Datasets	Evaluation technique	Evaluation metrics
[33]	Spectrogram	ViT (No details provided)	CHB-MIT	LOOCV	- Accuracy - Sensitivity - Specificity - Precision - AUC
[34]	CWT (scalograms)	Multi-channel ViT	-CHB–MIT -Kaggle/ American Epilepsy Society (AES) -Kaggle/ Melbourne University	LOOCV	- Accuracy - Sensitivity - Specificity - AUC - FPR (False Positive Rate)
[35]	Raw EEG	Temporal Multi-Channel Transformer (TMC-T) and ViT (TMC-ViT)	-CHB–MIT	5-fold CV (80% training and 20% testing)	- Accuracy - Sensitivity - Specificity - AUC
[36]	Utilizes multi-channel characteristics of EEG and Fourier transform	Convolutional transformer network with multiple gated inputs and Fourier transform (MFCvT)	-CHB-MIT -PKU1st dataset	–	- Accuracy - Sensitivity - Specificity - AUC

Table 2
Computational summary of the evaluated models.

Model	#Params	Size (MB)	Time/Fold (h)
CNN + Residual	6.09M	23.23	≈0.05
Mix-ViT (ResNet + ViT)	844M	321.80	≈0.10
Swin Transformer	86.75M	82.73	≈1.0

Computational setup

The transformer models were implemented using PyTorch, which required substantial GPU resources. Training and testing were performed on two platforms: Google Colab Pro with the A100 GPU, known for its high computational power, and a high-performance computer equipped with an Intel® Xeon® Silver 4110 CPU (16 cores @ 2.10 GHz) running Ubuntu 22.04.3 LTS. MATLAB was used for visualizing EEG signals, while Python libraries such as NumPy and Matplotlib supported data processing and visualization. Additionally, Stratified K-Fold cross-validation from `sklearn` was employed to ensure balanced class distribution across splits, improving the reliability of results.The quantitative computational summary, including parameter counts, model sizes, and average training times, is presented in Table 2, made against the regular CNN-based model for fair performance comparison.

Methods employed in the proposed system

This section provides an overview of important methods used in this research, offering insight into its subject matter. These terms encompass transformers, vision transformation, and the types of image representations employed.

Transformers

The Transformer, introduced in 2017, revolutionized the ability to capture long-range dependencies within datasets [37]. The transformer model architecture, comprising of encoder, decoder, and multi-head attention components, played a pivotal role in this development. Leveraging multi-head attention proves beneficial in capturing extended sequences, thereby enhancing overall results. The efficiency of transformers extends beyond Natural Language Processing (NLP), as demonstrated by Dosovitskiy et al.'s introduction of the ViT model for image recognition [31]. Their work highlights that convolutional networks can be replaced with pure transformers applied to sequences of image patches, achieving robust classification results. The study shows that ViT not only matches but often surpasses the performance of state-of-the-art convolutional networks. Additionally, ViT achieves this with significantly reduced computational requirements, particularly when pre-trained on large datasets and fine-tuned for mid-sized or small image recognition tasks. Fig. 2 illustrates the ViT architecture.

The ViT operates by first dividing an input image into a sequence of smaller, fixed-size patches. These patches are then flattened and linearly projected into a higher-dimensional space, which allows for the encoding of the patches as a series of vectors. Each patch is also associated with a unique position embedding to preserve the positional information, as the order of patches is essential for understanding the image content. An extra learnable embedding, often denoted as the [class] embedding, is prepended to the sequence of embedded patches. This [class] token evolves during the training process and serves as a representation of the entire image. The patch embeddings, along with the [class] embedding, pass through transformer encoder layers that use self-attention to focus on relevant image regions. The transformed [class] embedding is then processed by a Multi-Layer Perceptron (MLP) head, which predicts class probabilities. By treating images as sequences of patches, this architecture effectively applies transformers to image classification, achieving competitive performance with traditional CNNs on various benchmarks.

Mix vision transformer model

The advent of vision transformers has sparked numerous advancements in visual recognition tasks, with many models building upon this revolutionary architecture. Among them, the Mix-ViT model integrates mixing attentive transformers to capture fine-grained visual features, delivering strong performance across seven publicly available datasets for seizure classification [9]. This model's ability to learn from visual data is enhanced by combining ResNet-50's feature extraction with the sequential processing capabilities of transformer layers, making it a promising tool for identifying epileptic seizures patterns in images.

Fig. 3 illustrates the Mix-Vit model, which integrates ResNet-50 with transformer layers for visual data processing. The ResNet-50, a deep CNN with fifty layers, initially extracts features from the input images.

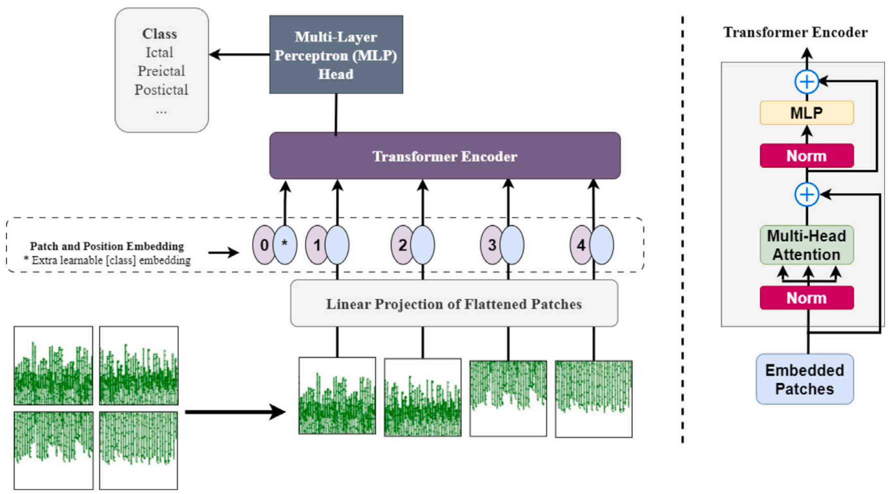

Fig. 2. ViT architecture.

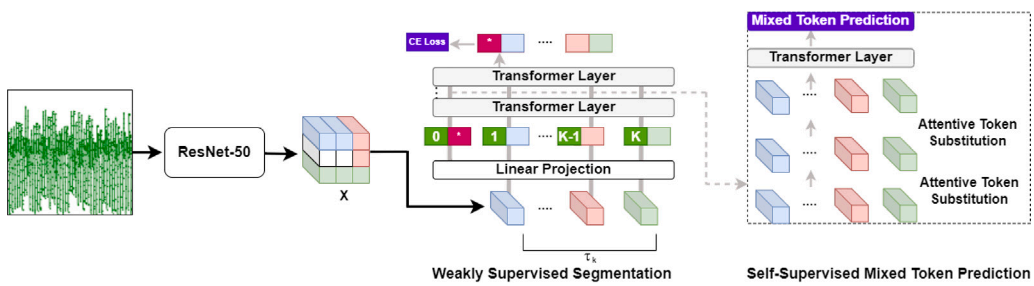

Fig. 3. Schematic representation of Mix-Vit model combining ResNet-50 with transformer layers.

These features are then processed by transformer layers to identify relationships within the image.

The features extracted by ResNet-50 undergo weakly supervised learning for part segmentation, with Cross-Entropy Loss used for optimization. The network extracts part-level features $\tau_k = (k = 1, 2, \ldots, K)$, where K represents the number of parts. Additionally, a self-supervised learning module with mixed token prediction improves the model's ability to autonomously generate labels, enhancing focus on important features through attentive token selection and substitution.

This architecture is applicable to various image-based recognition tasks, including seizure detection. By utilizing both the global context from transformer layers and localized features from ResNet-50, the model can detect complex patterns indicative of epileptic activity. This dual approach enables comprehensive analysis of the temporal and spatial aspects of brain imaging data, which is crucial for accurately predicting and detecting seizures.

Transformer layers were fine-tuned and adjusted, along with hyperparameter testing, to optimize model performance. Fine-tuning plays a key role in improving the results and adapting the model to the complexity of the problem.

SWIN Transformer model

The Swin Transformer represents a significant advancement in visual recognition architectures, diverging from traditional CNNs and building upon the principles of ViTs. Unlike conventional CNNs that rely on localized convolutional operations, Swin Transformer adopts a hierarchical transformer-based approach to process visual information efficiently. Its architecture breaks down input images into non-overlapping patches at multiple scales, enabling the model to capture both local and global context effectively. Notably, Swin Transformer introduces shifted windows, a departure from ViTs, which enhances its ability to capture spatial context without being limited by fixed-size receptive fields [38]. The comparison between Swin-ViT and ViT is shown in Fig. 4.

One of the key features of Swin Transformer is its hierarchical design, which facilitates the interaction between tokens at different levels, enabling the model to capture hierarchical representations of visual data. Additionally, Swin Transformer incorporates both local and global tokens, allowing it to simultaneously capture fine-grained details and global context. These features contribute to the model's ability to capture long-range dependencies across spatial dimensions in images, a crucial aspect for many visual recognition tasks. The Swin Transformer offers several advantages over traditional CNNs and ViTs. Its hierarchical architecture and shifted window mechanism enable it to efficiently capture long-range dependencies in images, leading to improved performance in various visual recognition tasks. Moreover, Swin Transformer exhibits scalability, allowing it to process high-resolution images without significantly increasing computational cost. Its modular design also facilitates easy integration with other architectures and enables seamless transfer learning across different domains and tasks.

Swin Transformer has found applications in a wide range of computer vision tasks, including image classification, object detection, and semantic segmentation. Its flexibility, scalability, and capability to capture long-range dependencies make it well-suited for addressing real-world challenges in computer vision applications. By leveraging its unique features, researchers and practitioners can achieve state-of-the-art performance in various visual recognition tasks.

Method for time series to image conversion

Transforming time series into images can unlock spatial features within seizure EEG signals, which may be pivotal in determining the presence of seizures. In this section, three specific image conversion

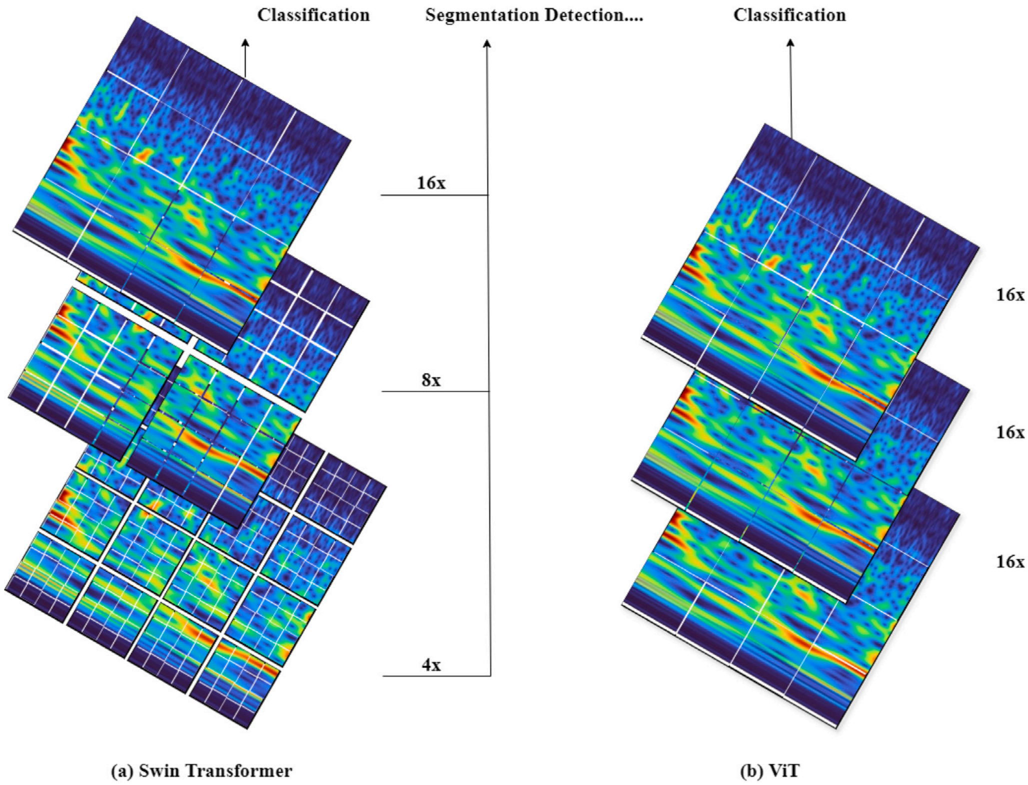

Fig. 4. Comparison between ViTs and Swin Transformers [38].

techniques that have demonstrated effectiveness in various fields, including seizure detection modeling, are explored for seizure detection. The transformation of time series data, initially presented as a text file containing a sequence of numerical values, into diverse graphical formats. Among these techniques are the spectrogram, which uncovers hidden correlations, and the scalogram, which applies wavelet transforms to reveal frequency-related characteristics [39]. Each technique is carefully selected to represent the unique features of EEG signals from seizures, to differentiate between seizure and non-seizure classes.

Raw image representation

Converting time series data directly into images is a straightforward and an effective approach that facilitates the use of image processing techniques on time-based signals. This direct transformation process leverages the inherent spatial relationships and patterns that may not be immediately apparent in the one-dimensional time series form. By translating these signals into a two-dimensional space, a variety of image analysis methodologies, such as CNN, which have been extensively refined and proven in the field of computer vision and seizure detection, can be applied [40]. This process effectively transforms the task of detecting complex temporal events, like seizures in EEG signals, into an image classification problem, where the rich and complex temporal dynamics are encapsulated as visual patterns. This technique bypasses the need for complex pre-processing steps, allowing for a more seamless and intuitive transition from raw data to analytical insights. A sample raw image representation from the CHB-MIT dataset and the Bonn University dataset is displayed in Fig. 5. It is noteworthy that eight channels from the CHB-MIT dataset were utilized, arranged in a 3 by 8 grid. This configuration was adopted to conform to the transformer model's requirement for a 256 by 256 image size, helping to mitigate any issues during the model's learning process.

Spectrogram

A spectrogram is a graphic representation of a signal's frequency spectrum that changes over time. It is made by taking the signal and using the STFT to visualize the frequency intensity with time. The constructed image is a two-dimensional graph where each point's color or intensity indicates the amplitude of a certain frequency at a given time, where one axis represents time, and another represents frequency. The spectrogram image's quality and accuracy are influenced by two crucial factors: the window's size and the quantity of overlapping dots.

$$x_1[n] = w[n] \cdot x[n + l.L], \quad 0 \le n \le N - 1 \tag{1}$$

Eq. (1) describes a step in the process of creating a spectrogram from a time series signal. It involves segmenting the original signal $x_1[n]$ into overlapping frames indexed by l. Each frame is then multiplied by a window function $w[n]$, which smooths the edges of the segment to minimize spectral leakage during the Fourier transform. The term $n+l \cdot L$ ensures that the windowing is applied to the correct segment in the sequence. Here, N represents the number of samples in each frame (window size), n is the starting time of window displacement, and L determines the overlap between consecutive frames [41]. This process prepares the signal for frequency analysis, where the Fourier transform of each windowed segment reveals its frequency content, ultimately contributing to the construction of the spectrogram.

The window length and overlap parameters were set in accordance with the methodology proposed [41]. For the healthy class, a window length of 2 s with an overlap of $L = 260$ was used, while for other classes, the window length was 1 s with an overlap of 131. The spectrograms derived from the CHB-MIT dataset and the Bonn University dataset are presented in Fig. 6.

Scalogram

A scalogram is an effective time–frequency transformation technique used for EEG signal analysis. The scalogram is essentially the absolute value of the CWT, graphed against time and frequency [42]. This representation provides a clear view of both the slowly varying and abrupt events in EEG waveforms. This method is particularly useful for

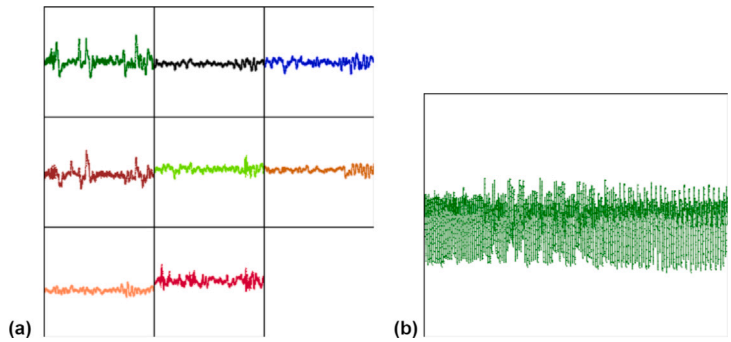

Fig. 5. (a) Raw representation for the CHB-MIT dataset for 8 EEG channels, and (b) raw representation for the Bonn University dataset for one channel Caption.

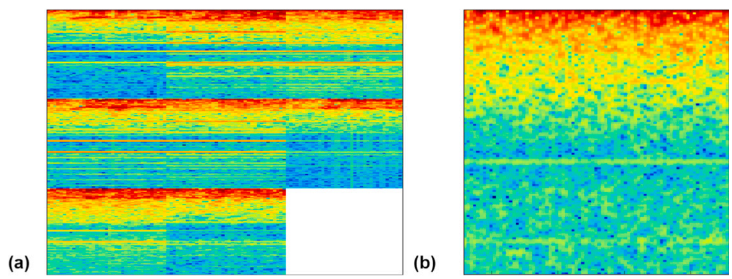

Fig. 6. (a) Spectrogram representation for the CHB-MIT dataset for 8 EEG channels, and (b) spectrogram representation for the Bonn University dataset for one channel.

classifying EEG data, as it allows for the analysis of the signal's time–frequency characteristics. This technique has shown significant promise in EEG data classification, offering a detailed representation of the signal's frequency–time characteristics. The scalograms derived from the CHB-MIT dataset and the Bonn University dataset are presented in Fig. 7.

Eq. (2) describes the CWT, which is used to analyze a signal into components of varying frequency and time. It works by shifting and scaling a wavelet function and comparing it to the signal. The translation parameter b shifts the wavelet along the time axis, while the scale parameter a adjusts its width, affecting the frequency resolution. This results in a two-dimensional analysis that reveals how the frequency content of the signal changes over time.

The equation for the scalogram, given the CWT of a signal, is straightforward. The scalogram is the squared magnitude of the CWT coefficients, represented as a function of time and scale given in Eq. (3).

$$CWT(b,a) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} x(t) h\left(\frac{t-b}{a}\right) dt \quad (2)$$

$$|S(b,a)| = |CWT(b,a)|^2 \quad (3)$$

Methodology for experimental evaluation

This section discusses the dataset used in our experiments (Section "Datasets"), the evaluation metrics employed (Section "Evaluation metrics"), and provides a detailed analysis of the results obtained using the two ViTs, comparing them with other state-of-the-art techniques (Section "Experimental results").

Datasets

In this paper, raw EEG datasets as sets of time series are utilized. There are many datasets available to train, evaluate and analyze the effectiveness of our model. In this article, two of the most commonly used datasets are discussed. These datasets are the Bonn University dataset and the CHB-MIT dataset [11,12]. It is important to highlight that no feature extraction techniques were applied prior to converting the dataset into image format. This article discusses these datasets in detail, providing more information about the used channels, sampling frequency, as well as the number of samples and what preprocessing steps were taken, if any.

Table 3
Bonn University dataset class set with description.

Set	Class
(A and B) or (Z and O)	Normal baseline or Surface
(C and D) or (N and F)	Preictal
E or S	Ictal

Bonn University dataset

Bonn university dataset is a publicly available dataset that is widely used in detecting seizure [11]. The dataset consists of three main classes such as preictal, ictal and interictal (as given in Table 3). The dataset size is 9.08 MB (9,523,339 bytes), which is small when compared to the other available datasets. The dataset consists of 5 main folders, each with 100 files, which means in total there are 500 files. Additionally, the dataset was sampled at 173.86 Hz sampling frequency in 23.6 s. The number of samples available per file accordingly is 4097. There were not any preprocessing steps made before converting the datasets into images. Every file with raw EEG signal data is processed through the designated graph function, resulting in its conversion into graphical form. Moreover, the full dataset was taken as one image without dividing it into time windows. Different combinations of sets and classes are tested and analyzed to ensure the effectiveness of the model on different sets of classes. The visualization of each set illustrated in Fig. 8.

CHB-MIT dataset

The dataset was produced by PhysioNet, which is a dependent open resource for recorded physiologic signals [12]. It was collected by Children's Hospital Boston (CHB) and Massachusetts Institute of Technology (MIT) in 2010. The CHB-MIT dataset is larger than the

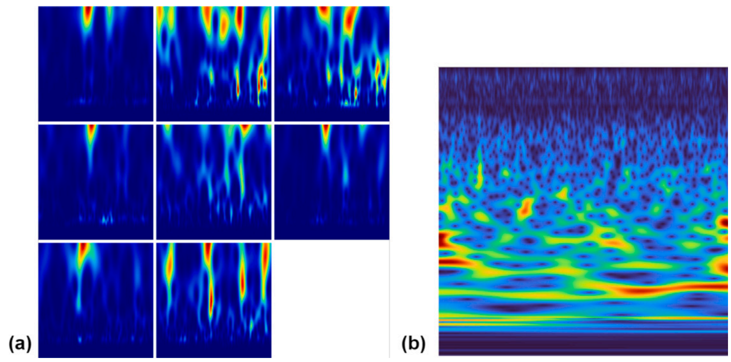

Fig. 7. (a) Scalogram representation for the CHB-MIT dataset for 8 EEG channels, and (b) scalogram representation for the Bonn University dataset for one channel.

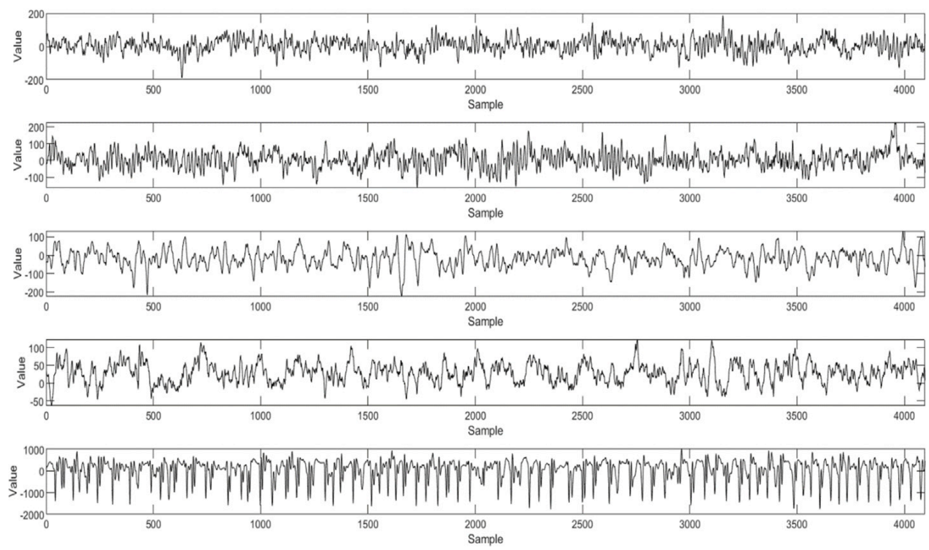

Fig. 8. The graph displays data from the Bonn University dataset, categorized into five classes (Z001, O001, N001, F001, and S001) from top to bottom.

Bonn University dataset, with a size of 42.6 GB, and it is widely utilized in many state-of-the-art techniques for detecting seizures. The dataset classifies binary classes of seizure and non-seizure. EEG signals were collected from 22 different subjects, resulting in 23 different cases since one subject's EEG signals were taken again after 1.5 years. Details of each case are illustrated in Table 4.

Additionally, the data are provided in EDF (European Data Format) files, which is a common format for biomedical signals and were sampled at 256 Hz. In this study, the following 8 channels ['FP1 - F7', 'P3-O1', 'P4-O2', 'FP2-F8', 'P8-O2', 'FZ-CZ', 'CZ-PZ', 'P7-T7'] were used for detecting seizures following the approach in [43]. The visualization of each channel is illustrated in Fig. 9.

This study employed the publicly available preprocessed and balanced CHB-MIT dataset [44], obtained from the IEEE DataPort. The dataset was derived from the original PhysioNet CHB-MIT EEG recordings. Preprocessing involved extracting balanced preictal and ictal data from the '.edf' files of 24 patients, each spanning approximately 68 min (4098 s) of EEG activity. Notably, only 10-second segments were used per sample, resulting in CSV files that are directly prepared for use in various machine and deep learning models. Each file maintains the standard 23 bipolar EEG channels recorded at a sampling rate of 256 Hz, with the final column indicating the class label (0 for non-seizure and 1 for seizure). This preprocessed version has also been utilized in follow-on studies such as [45], confirming its reliability and suitability for reproducible seizure detection experiments.

Evaluation metrics

Evaluation metrics are essential for assessing the performance of seizures detection models, ensuring accuracy, reliability, and comparability with state-of-the-art research. They highlight the strengths and weaknesses of models, guiding improvements in developing robust systems. This section covers key metrics like Accuracy, Sensitivity, Specificity, Precision, F1 Score, and Area Under the Curve (AUC), as shown in Eqs. (4)–(8), providing a comprehensive evaluation of model capabilities and performance.

There are some terminologies extensively used in defining evaluation metrics, as shown in Table 5.

Accuracy, represented in Eq. (4), is the proportion of TPs and TNs among all predictions, combining TP and TN divided by the sum of TP, FP, TN, and FN. Sensitivity (Eq. (5)), or Recall, measures the model's ability to correctly identify positive instances. Additionally, sensitivity is crucial as it measures the model's ability to correctly identify actual seizures, which is vital for minimizing missed seizure events. Likewise, specificity measures the proportion of the correctly predicted negative instances described by Eq. (6) [46].

The accuracy of positive prediction is calculated using precision, where it measures the total of positive prediction among all positive predictions given in Eq. (7). Moreover, the F1 Score, derived from Eqs. (5) and (7), is calculated by the harmonic mean of precision and recall given in Eq. (8). Furthermore, AUC, derived from the ROC (Receiver-Operating characteristic Curve), measures a model's ability

Table 4
Detailed explanation of the CHB-MIT dataset.

Case	Gender	Age	Number of files per case	Total seizure files
chb01	Female	11	50	7
chb02	Male	11	40	3
chb03	Female	14	46	7
chb04	Male	22	46	4
chb05	Female	7	45	5
chb06	Female	1.5	26	7
chb07	Female	14.5	23	3
chb08	Male	3.5	26	5
chb09	Female	10	23	3
chb10	Male	3	33	7
chb11	Female	12	39	3
chb12	Female	2	38	13
chb13	Female	3	42	8
chb14	Female	9	34	7
chb15	Male	16	55	14
chb16	Female	7	26	6
chb17	Female	12	25	3
chb18	Female	18	43	6
chb19	Female	19	34	3
chb20	Female	6	36	6
chb21	Female	13	38	4
chb22	Female	9	35	3
chb23	Female	6	13	3

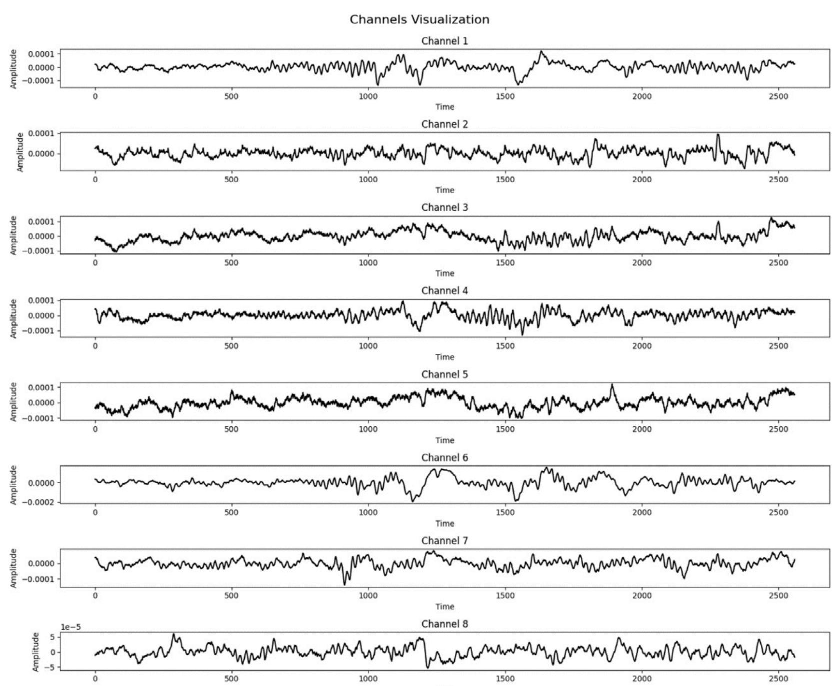

Fig. 9. CHB-MIT graphical representation of the 8 channels.

Table 5
Definitions of key evaluation metric terms.

Term	Definition
True Positive (TP)	Correctly predicted positive instances.
True Negative (TN)	Correctly predicted negative instances.
False Positive (FP)	Incorrectly predicted as positive.
False Negative (FN)	Incorrectly predicted as negative.

to distinguish between positive and negative classes, with higher values (ranging from 0 to 1) indicating better performance.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

$$\text{Sensitivity (Recall)} = \frac{TP}{TP + FN} \tag{5}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{6}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{7}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{8}$$

Experimental results

Various experiments were conducted using Mix-ViT and Swin architectures on the Bonn and CHB-MIT datasets, exploring different image transformations to capture diverse features. The models were implemented in Python with PyTorch, utilizing an A100 GPU from Colab Pro and other GPU-equipped computers for training. MATLAB

Table 6
Performance metrics of the Bonn University dataset utilizing Mix-ViT on five class classification.

Method	Acc.	Sens.	Spec.	Prec.	F1	AUC
Raw Images	0.754	0.67	0.917	0.666	0.667	0.795
Spectro.	0.882	0.876	0.96	0.879	0.87	0.922
Scalo.	0.856	0.844	0.961	0.841	0.841	0.902
Raw + Spectro.	0.994	0.98	0.994	0.980	0.97	0.987
Raw + Scalo.	0.991	0.963	0.988	0.948	0.951	0.976
Raw + Spectro. + Scalo.	**0.996**	**0.984**	**0.996**	**0.984**	**0.983**	**0.99**

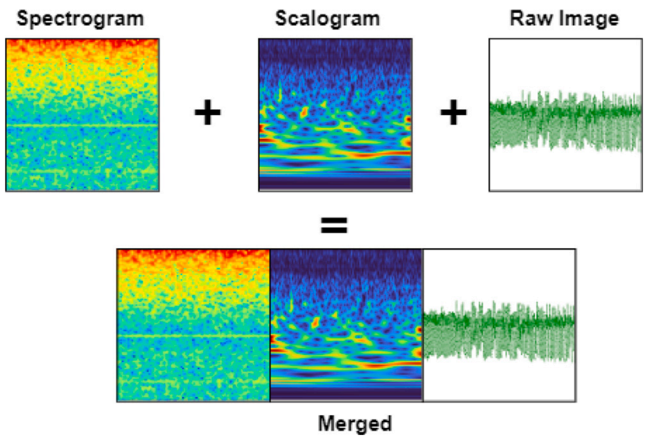

Fig. 10. A sample from the Bonn University dataset, merged image of three different image representations: raw images (left), spectrogram (center), and scalogram (right).

was used for plotting EEG signals, while libraries like NumPy and Matplotlib supported various functionalities. Stratified K-Fold cross-validation from `sklearn` ensured an equal distribution of classes across splits, promoting unbiased results.

The following section presents the results and analysis of the effects of different image combinations on two transformer models, Mix-ViT and Swin, applied to the Bonn University and CHB-MIT datasets.

Results on Mix-ViT for Bonn University dataset

Analyzing the Bonn University dataset's performance using Mix-ViT with various image combinations reveal clear trends (refer to Table 6). These results were obtained for 5-class classification, which is the most challenging scenario compared to other configurations. Raw images provide moderate accuracy and F1 scores, indicating limited feature representation. In contrast, spectrogram images significantly enhance performance across all metrics by effectively capturing discriminative information. Similarly, scalogram images also improve the results compared to using raw images alone.

Fig. 10 shows a sample of the merged input image used in the analysis. Combining raw images with either spectrogram or scalogram images results in substantial performance gains, with the raw + spectrogram combination demonstrating particularly high accuracy and precision. Leveraging all three image types (raw, spectrogram, and scalogram) together further boost performance, as illustrated in the accuracy and loss graphs (Fig. 11). These graphs show the results of the raw + spectrogram, raw + scalogram, and raw + spectrogram + scalogram combinations, emphasizing how combining spectrogram and scalogram images improves classification accuracy on the Bonn dataset using Mix-ViT. The smooth convergence of training and validation curves further confirms stable learning and the absence of overfitting with the combined representations.

After identifying that the combination of three image types produced the best results, comprehensive testing was conducted on the Bonn University dataset to evaluate the model's generalizability. Subsequent sections detail the testing of 23 different class combinations to ensure robust and reliable outcomes.

Results on binary classification

To ensure generalizability, additional experiments were conducted on binary classification, with extensive testing across 16 different class combinations. The results, presented in Table 7, demonstrate the exceptional performance of the Mix-ViT model in binary classification tasks across different class combination. The model achieved exceptional scores in accuracy, specificity, sensitivity, precision, F1 score, and AUC, consistently reaching 100% for most class combinations. To validate the reliability of the obtained results, the standard deviation across all folds was calculated. The outcomes revealed minimal variation, confirming the model's stable performance and consistent learning behavior across different training and testing splits. The minor deviations observed, such as 99.8% accuracy for the ABD-E class and 99.7% for the AB-CD class, are acceptable. These discrepancies stem partly from class imbalance, despite efforts to upsample class E using simple random duplication of minority class sample. Additionally, classifying the AB-CD classes poses challenges due to their similar features. On average, the model maintained 99.97% accuracy, underscoring its reliability and precision in accurately classifying diverse cases. This high level of performance across multiple metrics highlights the effectiveness of the Mix-ViT model in detecting and classifying epileptic seizures. This success is also attributed to the combination of multiple features in a single input image, allowing the model to leverage rich information for improved accuracy and generalization.

Results on three class classification

The model was also tested on three-class classification tasks, showcasing its effectiveness across various metrics. As detailed in Table 8, the results are exceptional for most class combinations, achieving perfect scores in accuracy, specificity, sensitivity, precision, F1 score, and AUC, all reaching 100% for combinations A-D-E, B-C-E, and AB-CD-E. These metrics indicate flawless classification performance, with no false positives or negatives. The A-C-E class metrics are nearly perfect, showing a slight drop in accuracy to 99.6%. However, the A-BC-DE combination exhibited lower performance, achieving 96% accuracy, 97.5% specificity, 95% sensitivity, 95.1% precision, 95% F1 score, and 96.25% AUC, which can be attributed to class imbalance. To address this, we implemented upsampling techniques to balance class A, aiming to enhance performance further.

While the model is highly effective overall, certain class combinations present more challenges. On average, the model demonstrates strong performance, with an accuracy of 99.08%, specificity of 99.5%, sensitivity of 99%, precision of 99.02%, F1 score of 99%, and AUC of 99.25%. In addition, the very low deviation values reported in Table 8 confirm the stability of the Mix-ViT model during training and testing, indicating that the performance is not influenced by data partitioning randomness or overfitting effects. This analysis highlights the model's robustness and reliability in classifying complex multi-class data.

Results on four and five class classification

The results shown in Table 9 reflect the application of Mix-ViT to the Bonn University dataset for both four-class and five-class classification tasks. For the four-class classification involving classes A, B, C, and E, the model excels across all metrics, achieving an accuracy of 99%, with specificity, sensitivity, precision, F1 score, and AUC all at 100%. In the five-class scenario, which includes an additional class D, the

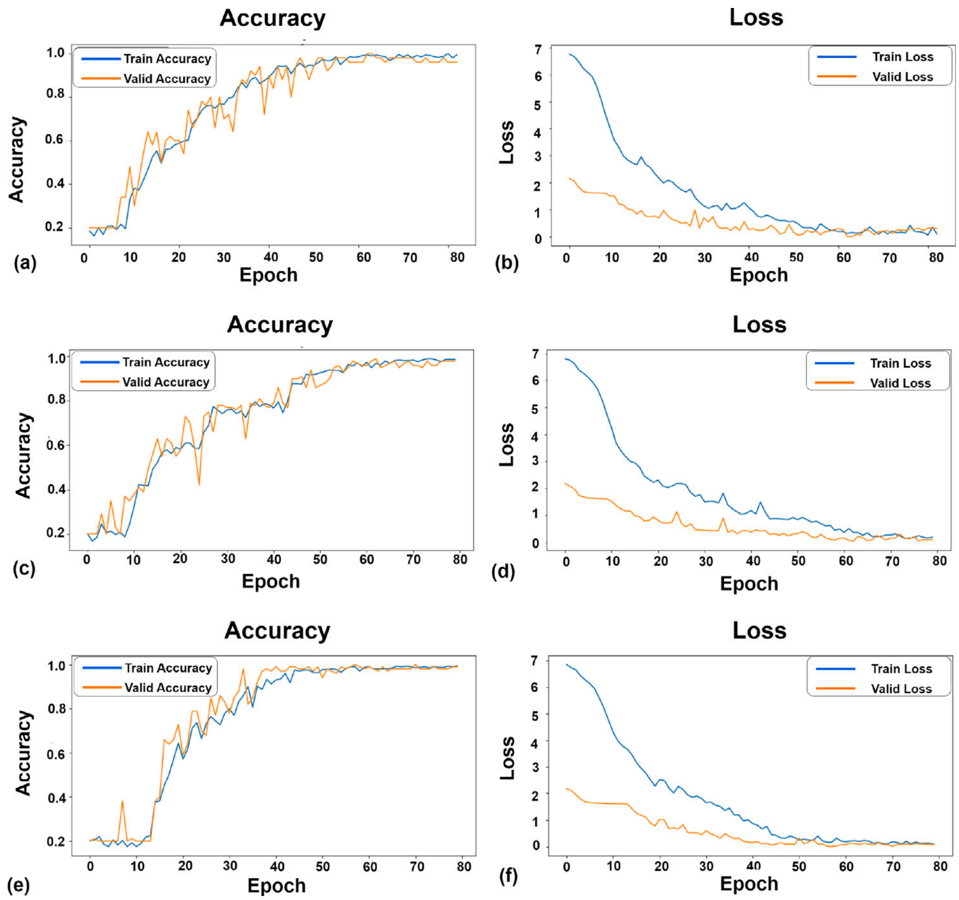

Fig. 11. The accuracy and loss graphs depict the performance of various image combinations trained on Mix-ViT. Image (a) and (b) illustrate the accuracy and loss, respectively, for the Raw + Scalogram images. Image (c) and (d) represent the same metrics for Raw + Spectrogram, while Image (e) and (f) display the results for Raw + Scalogram + Spectrogram.

Table 7
Results of merging three image representations on Bonn University dataset for binary classification using Mix-ViT (mean ± std across folds).

Classes	Accuracy (%)	Specificity (%)	Sensitivity (%)	Precision (%)	F1 score (%)	AUC (%)
A-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
A-B	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
B-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
C-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
D-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
AB-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
AC-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
AD-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
BC-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
CD-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
ABC-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
ABD-E	99.8 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
ACD-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
BCD-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
ABCD-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
AB-CD	99.7 ± 0.79	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
Avg.	**99.97**	**100 ± 0**	**100 ± 0**	**100 ± 0**	**100 ± 0**	**100 ± 0**

Table 8
Results of merging three image representations on Bonn University dataset for three-class classification using Mix-ViT (mean ± std across folds).

Classes	Accuracy (%)	Specificity (%)	Sensitivity (%)	Precision (%)	F1 score (%)	AUC (%)
A-D-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
A-C-E	99.6 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
B-C-E	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
A-BC-DE	99 ± 0.70	97.5 ± 0.44	95 ± 0.88	95.1 ± 0.84	95 ± 0.88	96.25 ± 0.66
AB-CD-E	99.8 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
Avg.	**99.08 ± 0.15**	**99.50 ± 0.36**	**99.00 ± 0.77**	**99.02 ± 0.64**	**99 ± 0.81**	**99.25 ± 0.63**

Table 9
Results of merging three image representations on Bonn University dataset for four and five class classification using Mix-ViT (mean ± std across folds).

Classes	Accuracy (%)	Specificity (%)	Sensitivity (%)	Precision (%)	F1 score (%)	AUC (%)
A-B-C-E	99 ± 0.9	100 ± 0	100 ± 0	100 ± 0	100 ± 0	100 ± 0
A-B-D-C-E	99.6 ± 0.9	98.4 ± 0.2	99.6 ± 1.3	98.4 ± 1.3	98.3 ± 1.2	99 ± 0.3

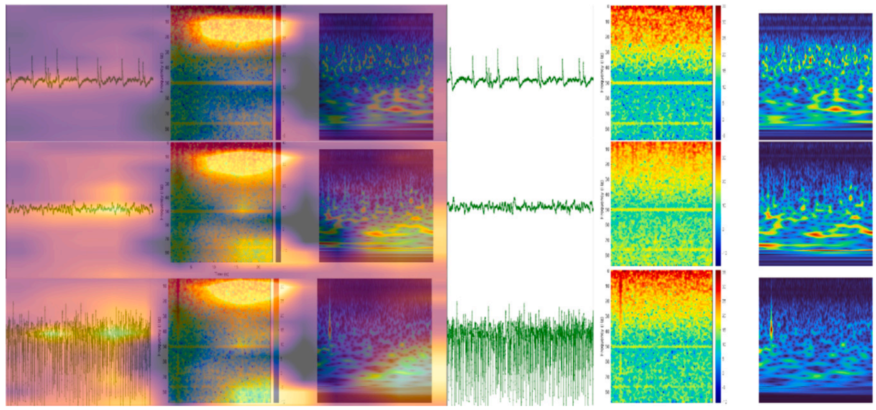

Fig. 12. Attention map visualizations of the Mix-ViT model on the Bonn EEG dataset for three representative classes (A: normal, C: pre-ictal, and E: seizure). Each row corresponds to one class, with the left section showing the Mix-ViT attention overlays on the fused EEG representations (raw, spectrogram, and scalogram), and the right section showing the corresponding original images.

model maintains impressive performance with an accuracy of 99.6%, specificity of 98.4%, sensitivity of 99.6%, precision of 98.4%, F1 score of 98.3%, and AUC of 99%. These results highlight the robustness of Mix-ViT in managing both four and five-class classification tasks on the Bonn University dataset. Despite the increased complexity with more classes, the model continues to yield fruitful outcomes, showcasing its effectiveness in various multi-class classification scenarios.

Attention map visualization and interpretability

To enhance the interpretability of the classification process, Fig. 12 presents the attention maps generated by the Mix-ViT model for three representative classes (A, C, and E) of the Bonn EEG dataset. Each row corresponds to one EEG class, where the left section displays the model's attention overlays on the fused EEG image representations (raw signal, spectrogram, and scalogram), and the right section shows the corresponding original images for comparison.

As illustrated, the model exhibits distinct attention behaviors for different physiological states. For the normal class (A), the attention is distributed more uniformly across the signal and frequency components, indicating stable, low-activity regions. For the pre-ictal class (C), the model begins to concentrate on moderate frequency fluctuations and transitional patterns, reflecting the onset of irregularities. In contrast, for the seizure class (E), the attention is strongly focused on dense and high-energy frequency bands, capturing the intense neural activity typical of seizure episodes. This progression highlights the model's ability to visually distinguish between normal, pre-ictal, and seizure EEG patterns.

Furthermore, these visualizations confirm the advantage of combining multiple EEG representations. The attention responses highlight complementary temporal and frequency-based information across the fused inputs, supporting the synergistic effect discussed in the results section. The observed performance gain therefore stems from the joint contribution of the raw EEG, spectrogram, and scalogram representations rather than from any single modality.

Results on SWIN transformer on Bonn University dataset

In the preceding section, an analysis of the Bonn dataset employing Mix-ViT was presented. Transitioning to the performance of the Swin Transformer on the same dataset, similar patterns are discerned across various image combinations (refer to Table 10). For five-class classification, raw images yield a moderate accuracy of 81.6%, indicating potential for enhancement in precision and F1 score. However, spectrogram and scalogram images achieve higher accuracies of 94% and 86%, respectively, with perfect sensitivity and specificity, thereby demonstrating effective feature representation. The combination of spectrogram and scalogram results in a slight accuracy decline to 72%, although sensitivity and specificity remain perfect. Integrating raw images with either spectrogram or scalogram significantly elevates performance to an accuracy of 98%. Furthermore, the utilization of all three image types results in perfect classification across all metrics (100%). These findings underscore the robustness of employing multiple image representations within the Mix-ViT and Swin Transformer models for five-class classification tasks, highlighting how multi-feature images enhance classification accuracy on the Bonn University dataset.

Results of Mix-ViT and SWIN transformer on CHB-MIT dataset

To ensure generalizability, the models were evaluated on the CHB-MIT dataset, which consists of two classes, in contrast to the Bonn University dataset, which comprises five classes. This testing involved both the Mix-ViT and SWIN Transformer models, utilizing various image combinations and only eight channels to assess their performance. Given the substantial size of the CHB-MIT dataset, testing was limited to evaluating the models on raw images and spectrograms to analyze the results effectively. The importance of utilizing multiple image representations as inputs to transformers cannot be overstated, as it significantly enhances the models' performance. As illustrated in Table 11, the Mix-ViT model achieved an accuracy of 87.9% with raw images alone and improved to 94.5% when combining raw data with spectrograms. This highlights how incorporating diverse representations can lead to more robust model performance.

In contrast, the SWIN Transformer model demonstrated even greater efficacy, reaching an accuracy of 96.34% and an AUC of 98.96% with the same combination. These findings underscore the transformative power of multi-image representations, indicating that leveraging varied input types not only enriches the data but also fosters improved classification capabilities. Ultimately, these results suggest promising avenues for further research and optimization in medical data analysis by integrating multi-representation data inputs.

Table 10
Performance of the swin transformer on the Bonn University dataset.

Method	Acc.	Sens.	Spec.	Prec.	F1	AUC
Raw Images	81.6	100	97	81.6	81.17	96
Spectro.	94	100	100	94.14	93.9	99.4
Scalo.	86	100	100	86	85.9	90.5
Spectro. + Scalo.	72	100	100	76	70.4	92.6
Raw + Spectro.	98	100	100	98.1	97.9	100
Raw + Scalo.	98	100	100	98.18	97.9	100
Raw + Spectro. + Scalo.	**100**	**100**	**100**	**100**	**100**	**100**

Table 11
Performance of the Mix-ViT and Swin Transformers on the CHB-MIT dataset.

Method	Acc.	Sens.	Spec.	Prec.	F1	AUC
Mix-ViT						
Raw Images	87.9	84.6	84.6	84.6	84.6	84.6
Spectro.	51.4	50	50	25	33	50
Raw + Spectro.	**94.5**	**93.8**	**93.8**	**93.9**	**93.8**	**93.8**
Swin Transformer						
Raw + Spectro.	**96.34**	**94.2**	**97.8**	**97.059**	**95.65**	**98.96**

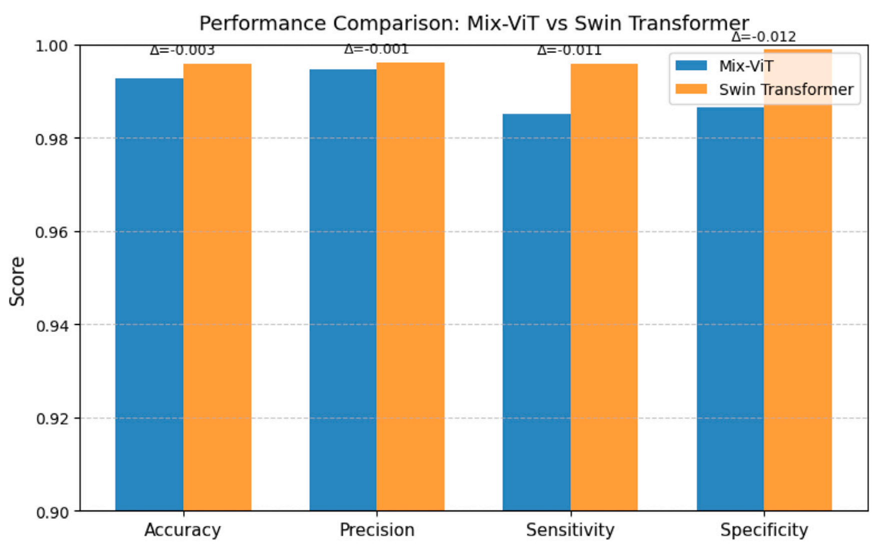

Fig. 13. Performance comparison between Mix-ViT and Swin Transformer across four key metrics using identical cross-validation splits.

Discussion

This section discusses the results, starting with a comparative statistical analysis of the Swin-ViT and Mix-ViT models (Section "Comparative statistical analysis: Mix-ViT vs Swin-ViT"). For the Bonn University dataset, the models are evaluated against recent state-of-the-art methods across binary, three-, and five-class tasks (Sections " Binary classification on the Bonn University dataset"–"Five class classification on the Bonn University"). Finally, CHB-MIT results are compared with other studies to confirm the robustness of the proposed models (Section "Binary classification on the CHB-MIT dataset").

Comparative statistical analysis: Mix-ViT vs Swin-ViT

To quantitatively evaluate the comparative performance between Swin-ViT and Mix-ViT, a paired t-test was performed using identical cross-validation splits across ten folds of the Bonn University dataset. The analysis confirmed that Swin-ViT achieved consistently higher mean values across all evaluation metrics, with a mean accuracy difference of $\Delta = +0.01$ in favor of Swin-ViT ($t = 1.15$, $p = 0.28$, Cohen's $d = 0.36$). Although the differences were not statistically significant ($p > 0.05$), the positive effect size and consistent direction of improvement indicate a measurable advantage of Swin-ViT in generalization and stability across folds.

Fig. 13 further supports these observations, illustrating Swin-ViT's superior discriminative capacity across all evaluation metrics. While both models achieved near-perfect accuracy, Swin-ViT maintained slightly higher precision, sensitivity, and specificity, confirming its robustness and stronger generalization in multi-class EEG classification.

Binary classification on the Bonn University dataset

For binary classification using Mix-ViT, 16 different class combinations were tested on the Bonn University dataset. As the only model tested across various class configurations, Mix-ViT consistently demonstrated superior performance, achieving 100% accuracy, specificity, and sensitivity in most cases (as shown in Table 12). These results emphasize the benefits of mixed feature combination for improving performance on smaller datasets. Despite prior concerns about Vision Transformers' performance on small datasets [31,32], our results confirm that Mix-ViT can achieve state-of-the-art accuracy across multiple scenarios.

In class combinations such as A-E, B-E, C-E, AB-E, AC-E, AD-E, BC-E, CD-E, and ABC-E, Mix-ViT achieved perfect scores of 100% in all evaluation metrics, surpassing or matching the results of prior studies, including those by Abualhassan et al. [46], Zeng et al. [8], and Huang et al. [27]. In the D-E combination, where Kavitha et al. [16] reported 93.5% accuracy, Mix-ViT significantly outperformed with a perfect

Table 12
Results of the Mix-Vit model versus recent state-of-the-art papers on binary classification of Bonn University dataset.

Classes	Ref.	Accuracy (%)	Specificity (%)	Sensitivity (%)
A-E	Kavitha et al. [16]	100	100	100
	Huang et al. [47]	100	100	100
	Abualhassan et al. [46]	99.5	–	–
	Mix-ViT	**100**	**100**	**100**
A-B	Abualhassan et al. [46]	82.5	–	–
	Mix-ViT	**100**	**100**	**100**
B-E	Kavitha et al. [16]	99.5	99	100
	Huang et al. [47]	100	100	100
	Abualhassan et al. [46]	98.5	–	–
	Mix-ViT	**100**	**100**	**100**
C-E	Kavitha et al. [16]	98	97	99
	Zeng et al. [8]	100	–	–
	Huang et al. [47]	100	100	100
	Abualhassan et al. [46]	98.5	–	–
	Mix-ViT	**100**	**100**	**100**
D-E	Kavitha et al. [16]	93.5	97	90
	Zeng et al. [8]	100	–	–
	Huang et al. [47]	100	100	100
	Abualhassan et al. [46]	99.5	–	–
	Mix-ViT	**100**	**100**	**100**
AB-E	Kavitha et al. [16]	99.67	100	99.5
	Huang et al. [47]	100	100	100
	Abualhassan et al. [46]	99.34	–	–
	Mix-ViT	**100**	**100**	**100**
AC-E	Kavitha et al. [16]	98.66	97	99.5
	Mix-ViT	**100**	**100**	**100**
AD-E	Kavitha et al.	96.66	96	96
	Mix-ViT	**100**	**100**	**100**
BC-E	Kavitha et al. [16]	98.6	97	99.5
	Mix-ViT	**100**	**100**	**100**
CD-E	Kavitha et al.	96.6	93	98.5
	Mix-ViT	**100**	**100**	**100**
ABC-E	Kavitha et al. [16]	99	97	99.6
	Mix-ViT	**100**	**100**	**100**
ABD-E	Kavitha et al. [16]	98.25	96	99
	Mix-ViT	**99.8**	**100**	**100**
ACD-E	Kavitha et al. [16]	97.5	96	98
	Mix-ViT	**100**	**100**	**100**
BCD-E	Kavitha et al. [16]	100	100	100
	Mix-ViT	**100**	**100**	**100**
ABCD-E	Kavitha et al. [16]	97.4	97	97.5
	Huang et al. [47]	100	100	100
	Abualhassan et al. [46]	99.3	–	–
	Mix-ViT	**100**	**100**	**100**
AB-CD	Kavitha et al. [16]	72.5	71	74
	Abualhassan et al. [46]	97.25	–	–
	Mix-ViT	**99.7**	**100**	**100**

score of 100%. For the ABD-E class combination, Mix-ViT achieved 99.8% accuracy, slightly below Kavitha et al.'s 98.25%, but reached 100% specificity and sensitivity, outperforming previous methods.

Moreover, the use of mixed image representations as input had a significant effect on enhancing the results. Mix-ViT achieved an impressive 99.7% accuracy for the challenging AB-CD class combination, marking a substantial improvement over Kavitha et al.'s reported 72.5%. These results underscore Mix-ViT's robustness in identifying seizures across various binary classification tasks, surpassing or matching the performance of other state-of-the-art approaches.

Three class classification on the Bonn University

For three-class classification, Mix-ViT demonstrates outstanding performance, achieving nearly 100% accuracy, specificity, and sensitivity across different class combinations, matching the results reported by Zeng et al. [8] and Islam et al. [48], as shown in Table 13. While Zeng et al. used Gray Recurrence Plot (GRP) to transform EEG signals into images, our model achieved similar outcomes with combined representation images. Although Zeng et al. did not report specificity and sensitivity, Mix-ViT showed excellent results across all metrics.

For the A-C-E combination, Mix-ViT achieved a high accuracy of 99.6%, comparable to the results of Islam et al. [48] and surpassing those reported by Abualhassan et al. [46]. Notably, our model performed well without any preprocessing of the input images, unlike Islam et al. who applied band-pass filtering and non-overlapping window splitting. This underscores the effectiveness of our approach in converting full-time series data into images and leveraging ViT.

In the B-C-E combination, Mix-ViT once again achieved perfect scores, outperforming previous studies, including Abualhassan et al. [46]. Overall, the results are impressive, particularly given the small dataset used for Vision Transformers, highlighting Mix-ViT's capability in handling three-class classification tasks effectively.

Table 13
Results of the Mix-ViT model versus recent state-of-the-art papers on three-class classification of the Bonn University dataset.

Classes	Ref.	Accuracy (%)	Specificity (%)	Sensitivity (%)
A-D-E	Zeng et al. [8]	100	–	–
	Abualhassan et al. [46]	94.7	–	–
	Mix-ViT	**100**	**100**	**100**
A-C-E	Islam et al. [48]	100	–	–
	Abualhassan et al. [46]	92	–	–
	Mix-ViT	**99.6**	**100**	**100**
B-C-E	Abualhassan et al. [46]	96	–	–
	Mix-ViT	**100**	**100**	**100**
AB-CD-E	Zeng et al. [8]	100	–	–
	Abualhassan et al. [46]	97.6	–	–
	Mix-ViT	**99.8**	**100**	**100**
A-BC-DE	–	–	–	–
	Mix-ViT	**100**	**100**	**100**

Table 14
Comparison of classification models on 5 classes.

Classes	Ref.	Accuracy (%)	Specificity (%)	Sensitivity (%)
Five class classification	Do and Le [49]	99.72	100	100
	Islam et al. [48]	99.96	–	–
	Abualhassan et al. [46]	88.25	–	–
	Mix-ViT	**99.6**	**99.6**	**96.3**
	Swin-ViT	**100**	**100**	**100**

Five class classification on the Bonn University

In the five-class classification task, Swin-ViT achieved outstanding results, with 100% accuracy, specificity, and sensitivity across all metrics (refer Table 14). This performance surpasses the 99.72% accuracy reported by Do and Le [49] and outperforms the 99.96% accuracy noted in Islam et al. [48], which did not reach perfect scores across all evaluation metrics. Additionally, the results for Mix-ViT, while slightly lower, remain highly competitive, achieving 99.6% accuracy, 99.6% specificity, and 96.3% sensitivity. The robustness of Swin-ViT, particularly in achieving flawless classification, emphasizes its effectiveness in handling complex multi-class tasks, even with the challenges posed by small datasets and imbalanced sample proportions across classes. This achievement highlights Swin-ViT's advanced capacity to capture and classify intricate patterns, setting a high standard in seizure classification.

Binary classification on the CHB-MIT dataset

In the five-class classification task, Swin-ViT achieved outstanding results, with 100% accuracy, specificity, and sensitivity across all metrics. This performance surpasses the 99.72% accuracy reported by Do and Le [49] and outperforms the 99.96% accuracy noted in Islam et al. [48], which did not reach perfect scores across all evaluation metrics. Additionally, the results for Mix-ViT, while slightly lower, remain highly competitive, achieving 99.6% accuracy, 99.6% specificity, and 96.3% sensitivity. The robustness of Swin-ViT, particularly in achieving flawless classification, emphasizes its effectiveness in handling complex multi-class tasks, even with the challenges posed by small datasets and imbalanced sample proportions across classes. This achievement highlights Swin-ViT's advanced capacity to capture and classify intricate patterns, setting a high standard in seizure classification.

The Swin-ViT also outperformed several other studies [18,19,22,28,33,50,51]. For example, compared to Usman et al. [18], Swin-ViT achieved a 1.2% higher sensitivity (94.2% vs. 93%) and a 6.07% improvement in specificity (97.8% vs. 92.2%). Similarly, against Yang et al. [51], it showed a 4.6% increase in accuracy and an 9.5% gain in specificity. Despite using fewer channels (8 versus 22), Mix-ViT reached a sensitivity of 93.8%, which was comparable to Yang et al.'s results, demonstrating the models' efficiency. It is worth noting that Yang et al. employed LOOCV for their evaluation, which could further enhance their results, in contrast to the 10-fold cross-validation method used in this paper. Although our performance compared to Li et al. [47] who utilized transformers, showed only marginal improvement of 0.7% in sensitivity, these notable enhancements underscore the efficacy and superiority of the Swin-ViT model over the methodologies proposed in these studies.

The Swin-ViT further outperformed Varlıand Yılmaz [28] by 1.32% in accuracy and Assali et al. [17] by 6.9%, underscoring its competitive advantage in EEG-based seizure detection. While Hussein et al. [34] achieved a higher accuracy of 99.8%, Swin-ViT still demonstrated strong performance with 96.34% accuracy, 94.2% sensitivity, and 97.8% specificity. Despite Hussein et al.'s impressive accuracy of 99.8%, sensitivity of 99.8%, and specificity of 99.7%, our Swin-ViT model exhibits a respectable performance. Notably, while Hussein et al. utilized a more complex approach with scalograms and 23 channels, our model achieved comparable results through a simpler methodology. This suggests the effectiveness and efficiency of our Swin-ViT model in capturing relevant features and detecting seizures, utilizing multiple image representations while requiring fewer channels and less computational intensity.

Overall, the comparative analysis highlights the competitive nature of the proposed models, with Swin-ViT particularly standing out as a robust method for seizure prediction using the CHB-MIT dataset. Table 15 presents a comparison of the results with other state-of-the-art studies on CHB-MIT dataset.

Conclusion

Epileptic seizure detection is a challenging problem that varies significantly between patients. Capturing key features is vital for accurate detection, while reducing input channels significantly lowers computational demands. In this paper, three key image representations – spectrogram, scalogram, and raw images – were analyzed on two publicly available datasets: the Bonn University dataset and the CHB-MIT dataset.

Results showed that combining multiple image representations improved performance, often surpassing single-representation results, as the attention maps revealed how the models effectively concentrated on the most informative temporal and frequency regions across the EEG images. The Bonn University dataset achieved nearly 100% accuracy across all evaluation metrics in most test cases. For the CHB-MIT

Table 15
Comparison of accuracy, specificity, and sensitivity from various studies.

Reference	Accuracy (%)	Specificity (%)	Sensitivity (%)
Assali et al. [17]	90.1	–	92.3
Baghdadi et al. [50]	96.48	–	–
Zhao et al. [7]	98.76	97.60	97.70
Varlı & Yılmaz [28]	95.08	93.28	96.7
Zhang & Li [33]	81.1	81.7	–
Yang et al. [51]	92.07	89.25	92.67
Li et al. [47]	–	–	93.5
Hussein et al. [34]	99.8	99.7	99.8
Usman et al. [19]	–	92.5	93
Usman et al. [18]	–	92.2	93
Mix-ViT	**94.5**	**93.8**	**93.8**
Swin-ViT	**96.34**	**97.8**	**94.2**

dataset, only eight channels were utilized, and the results were comparable to those obtained with a higher number of channels, demonstrating the effectiveness of reducing input dimensions while maintaining high performance.

Extensive testing was conducted to validate the effectiveness of combining these image representations, with approximately 23 different combinations tested and analyzed for the Bonn dataset. Paired t-test evaluations further confirmed the strong performance of both Mix-ViT and Swin-ViT models, with Swin-ViT exhibiting a slight yet consistent advantage in accuracy and stability across all evaluation metrics. Additionally, the use of two recent vision transformer architectures demonstrated substantial improvements compared to other existing approaches, highlighting the importance of integrating multiple EEG image representations to capture diverse and complementary features for more comprehensive seizure detection.

In conclusion, this study opens up exciting possibilities for future research in epileptic seizure detection. One promising direction is the integration of diverse modalities, which could further enhance model performance by utilizing the unique features each representation offers. By integrating these diverse modalities, a more comprehensive understanding of seizure activity can be achieved. Additionally, employing faster exploratory methods to analyze these representations could streamline the evaluation process, enabling researchers to quickly assess their effects on model efficacy. To ensure the model's generalizability, applying this approach to larger datasets is essential, as it would confirm its robustness across a broader range of patient data. This strategy not only aims to refine detection capabilities but also holds the potential for creating more personalized systems that serve to individual patient needs, ultimately improving clinical outcomes in seizure management.

CRediT authorship contribution statement

Zainab Abualhassan: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Imtiaz Ahmad:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Conceptualization. **Sa'ed Abed:** Writing – review & editing, Writing – original draft, Validation, Formal analysis. **Ehtesham Hassan:** Writing – review & editing, Writing – original draft, Co-Supervision, Validation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Not applicable.

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